Reference Books

1. Heywood, J. B., Internal Combustion Engine Fundamentals, McGraw Hill, New York (1988).

Evaluation Scheme

S.No.	Evaluation Elements	Weightage(%)
1.	MST	30
2.	EST	45
3.	Sessionals (Assignments/Tutorials/Quizes)	25

UME749 : Gas Turbine Theory and Design

L	T	P	Cr
3	0	0	3.0

Course Objective: Students will be exposed to basics of energy systems, heat transfer and thermodynamics applied to gas turbines and compressors. In addition, the student will learn about gas turbine cycle and its modifications, three-dimensional flows in turbo machines, design of individual components, and the prediction of designed and off-design performance, blade materials, blade attachments and cooling, plant performance and matching, applications of gas turbine power plants.

Syllabus

Review: Development, classification and field of application of gas turbines, Gas turbine cycle, Multistage compression, Reheating, Regeneration combined and cogeneration, Energy transfer between fluid and rotor, Axi-symmetric flow in compressors and gas turbines.

Compressors: Classification, Centrifugal compressors, Adiabatic efficiency, Slip factor, Design consideration for impeller and diffuser systems, Performance characteristics, Axial flow compressors, Vortex theory, Degree of reaction, Simple design, Aerofoil theory, Cascade theory, Stages, Stage efficiency and overall efficiency, performance characteristics. Combustion systems, Design considerations, Flame stabilization

Turbines: Classification - axial flow and radial flow turbines, Elementary vortex theory, Aerodynamic and thermodynamic design considerations, Blade materials, Blade attachments and cooling, Gas turbine power plants, Plant performance and matching, Applications of gas turbine power plants.

Fans and Blowers: Fan applications, Types, Fan stage parameters and design.

Course Learning Objectives (CLO)

The students will be able to:

1. Analyse and design centrifugal compressor.
2. Analyse and design axial flow compressor stages.
3. Analyse and design axial and radial flow gas turbine.
4. Design for matching of the components of gas turbine power plant.

Recommended Books

1. Cohen, H., Rogers, G.F.C., and Saravanamuttoo, H.I.H., Gas Turbine Theory, Longman (2008).
2. Oates, G.C., Aero-thermodynamics of Gas Turbine and Rocket Propulsion AIAA Education Series (1997).
3. Yahya, S.M, Turbines, Compressors and Fans, Tata McGraw-Hill (2005).
4. Dixon, S.L., Fluid Mechanics and Thermodynamics of Turbomachinery, Elsevier.
5. Ganesan, V., Gas Turbines, Tata McGraw-Hill (2003)

Evaluation Scheme

S. No.	Evaluation Elements	Weightage (%)
1.	MST	30
2.	EST	45
3.	Sessional (Assignments/Projects/Tutorials/Quizes)	25